

HEOR Flagship Project Deep Research

Executive Judgment The governing strategy for this project should remain the ChatGPT framing: one repo, T2DM as the flagship disease area, Phase 1 limited to a reproducible synthetic cohort engine and poster-ready evidence package, propensity methods deferred to Phase 2, and payer-facing economics deferred to a later phase. That framing is superior because it defines what "done" means for Year 1 with unusual clarity: a locked study protocol, reproducible cohort counts and diagram, a baseline table and treatment-pattern figure, one survival or utilization figure plus a cost summary, and a limitations section stating clearly that the data are synthetic. That is the actual definition of a June 2027 faculty-ready poster.

The clinical use case should also remain T2DM, not AFib. For a managed care and industry HEOR trajectory, the live formulary and pharmacoeconomic conversation is in metabolic disease: SGLT2 inhibitors, GLP-1 receptor agonists, cardiovascular and CKD protection, budget impact, and value. AFib DOACs versus warfarin is a clean teaching case, but it is a more settled question and is strategically weaker for fellowship signaling.

The most important scope decision is timing. Phase 1 only should happen before school starts. That means cohort extraction, attrition logic, baseline table, Parquet normalization, and the OMOP-lite SQLite layer. Phase 2 begins later, in the two-week breaks between sessions and during Session 2, when propensity methods, Kaplan-Meier output, and any exploratory Cox modeling can be added without setting up failure during the first academic term.

Why HEOR First in Practical Terms HEOR is the right first build because it matches both your constraints and your comparative advantage. You already have software architecture depth; what you need now is visible training in study design, cohort logic, temporal assembly, reporting discipline, and payer-oriented evidence framing. Synthea gives you a privacy-free longitudinal environment where those muscles can be built without IRB delay, DUAs, or brittle access dependencies.

This project should be presented as a methods-development complement to retrospective oncology work, not a substitute for real RWD. The real value is learning how to pre-specify eligibility rules, define index dates, assemble baseline windows, generate attrition logic, control provenance, and produce reproducible artifacts before touching protected data. That is the transferable skill stack faculty, informatics collaborators, and managed care evaluators will respect.

Recommended Flagship Project Definition The exact Phase 1 project is this: A reproducible synthetic real-world evidence pipeline for adult type 2 diabetes that ingests Synthea longitudinal records, constructs a pre-specified incident T2DM cohort, outputs baseline and follow-up analytic datasets, characterizes treatment and monitoring patterns, estimates one time-to-event or utilization outcome, and generates a parameterized report with transparent synthetic-cost summaries.

The first study design should be a retrospective longitudinal incident-cohort study in synthetic data, not an active-comparator causal study. Define the population as adults aged 18 years or older with a first qualifying T2DM diagnosis after a baseline observation window. Require at least 365 days of observable history before index and at least one diabetes-relevant lab or monitoring event in the surrounding period if available. Exclude obvious non-target phenotypes such as type 1 diabetes and gestational or pregnancy-associated diabetes windows where the record structure makes them identifiable.

The cohort logic should be shown explicitly rather than left as prose:

Cohort Selection Pipeline

Raw Synthea Population (e.g., $n = 100,000$)

↓

Inclusion Criterion 1: Diagnosis of T2DM (SNOMED 44054006)

- No → Exclude
- Yes → Continue

↓

Inclusion Criterion 2: Age ≥ 18 years at index date

- No → Exclude
- Yes → Continue

↓

Inclusion Criterion 3: First qualifying T2DM diagnosis occurs after the baseline observation window

- No → Exclude
- Yes → Continue

↓

Inclusion Criterion 4: ≥ 365 days of observable history before index date

- No → Exclude
- Yes → Continue

↓

Inclusion Criterion 5: Diabetes-relevant monitoring or laboratory context available, if applicable

- No → Exclude
- Yes → Continue

↓

Exclusion Criterion 1: Prior type 1 diabetes diagnosis (SNOMED 46635009)

- Yes → Exclude
- No → Continue

↓

Exclusion Criterion 2: Pregnancy or gestational diabetes window (SNOMED 77386006)

- Yes → Exclude
- No → Continue

↓

Final Analytical Cohort

The vocabulary layer should also be concrete from the start:

Concept	Vocabulary	Code(s)	Use
Type 2 Diabetes Mellitus	SNOMED-CT	44054006	Target condition
Type 1 Diabetes Mellitus	SNOMED-CT	46635009	Exclusion
Pregnancy	SNOMED-CT	77386006	Exclusion or gestational-window control
Heart failure hospitalization	SNOMED-CT	85232005	Possible event component
Myocardial infarction	SNOMED-CT	57054005	Possible MACE component
Ischemic stroke	SNOMED-CT	230690007	Possible MACE component
Metformin	RxNorm	860974	Baseline treatment context
SGLT2 inhibitors	RxNorm	1537213, 1373456	Phase 1 treatment-pattern analysis
GLP-1 receptor agonists	RxNorm	1991302, 1151131	Phase 1 treatment-pattern analysis

The core input tables for the MVP should be the clinical minimum needed for serious cohort work: patients, encounters, conditions, medications, observations, procedures, and careplans. Claims-like tables, payer/provider tables, or CPCDS-style exports are useful but should be treated as optional MVP-plus inputs rather than blockers.

The index date should be the first qualifying T2DM diagnosis that meets the adult-age and baseline-history rules. Follow-up should run for up to 24 months or until death or end of record, whichever comes first. The primary outputs should be a cohort attrition table, a baseline characteristics table, a medication-exposure summary by class, a lab-monitoring summary, one longitudinal A1c or glucose trajectory view where available, one Kaplan-Meier analysis for a clearly defined event, one utilization summary, and one transparent synthetic-cost summary by outcome category.

The realistic Phase 1 analyses are descriptive statistics, cohort attrition, standardized mean differences for descriptive subgroup balance, Kaplan-Meier curves, log-rank tests, and at most one exploratory Cox model with careful non-causal language. What is too ambitious for the first build is recurrent-event modeling, multistate modeling, comparative-effectiveness claims, recurrent-cost microsimulation, full probabilistic sensitivity analysis, or a dashboard-first delivery layer.

Phase Architecture and Technical Blueprint This should be one coherent repo with staged capabilities. The best sequence is: Phase 1 cohort engine and descriptive outcomes, Phase 2 comparative-methods sandbox, Phase 3 payer-facing economic layer, and Phase 4 optional reporting interface. The base strategy remains ChatGPT's scope discipline, but the technical framing should absorb two key additions: an OMOP-lite mapping layer in Phase 1 and concrete module interfaces that make the repo immediately scaffoldable.

Phase 1 should include an OMOP-lite mapping layer from the start, but not a full institutional OMOP implementation. The right compromise is to ingest raw Synthea CSVs, normalize them to Parquet, and map the core entities into a lightweight local SQLite relational structure aligned to familiar OMOP concepts such as person, visit_occurrence, condition_occurrence, drug_exposure, observation, and procedure_occurrence. That gives you immediate industry signaling and a place for SQL extraction scripts without turning the MVP into a full CDM conversion project.

Phase 1 is the build you should aim to have poster-ready by the end of PY1. The endpoint is a disease-specific cohort extraction engine with deterministic eligibility logic, analytic dataset creation, descriptive treatment and monitoring summaries, one survival endpoint, one utilization and cost summary, reproducible cohort counts, and a fully reproducible report.

Phase 2 adds comparative-methods training: active-comparator or new-user cohort logic, PS matching or IPTW, target-trial-style protocol framing, optional synthetic confounding injection, and optional medication diversification. The key justification is important: PSM belongs in Phase 2 not because default Synthea requires it for validity, but because proving methodological competence to human evaluators is the whole point. The repository should explicitly show logistic-regression propensity estimation, 1:1 nearest-neighbor matching with a prespecified caliper, and standardized mean difference checks using the usual balance formula: $SMD = (x_1 - x_0) / \sqrt{(s_1^2 + s_0^2) / 2}$. Because default Synthea does not generate organic confounding-by-indication the way real claims data does, the matched cohort output can be checked against known ground-truth distributions, making this a better PSM training environment than messy real data for a first build.

The documentation should still say clearly that the exercise demonstrates method execution in a synthetic sandbox rather than genuine confounding-by-indication correction.

Phase 3 is where the earlier synthetic claims cost-effectiveness analyzer should live. It belongs in the same repo, but not in the MVP. The mathematical bridge between Phase 2 and Phase 3 should be explicit: the Markov cohort trace evolves as $S_{(t+1)} = S_t \times P$, and the economic comparison is summarized as $ICER = \Delta C / \Delta E$. That is the formal link between survival-derived parameters from Phase 2 and transition probabilities in the later payer-facing model.

Phase 4 should be optional and light. If you ever add it, make it a read-only reporting layer, not an application platform. A parameterized Quarto report or a very small Streamlit front end is enough. For PY1 goals, a high-quality report generator is worth more than an interactive dashboard.

The technical stack should be opinionated but compact. Use Polars as the primary data layer. Use Parquet as the canonical analytical store. Use DuckDB for ad hoc SQL, QA, and validation queries over Parquet. Use SQLite as the local OMOP-lite relational layer, run registry, and manifest store. SQLite should not become a giant institutional warehouse, but it is worth using in Phase 1 because SQL extraction scripts are part of the signal you want this repo to send.

The repo architecture should keep definitions declarative and analyses modular:

```
heor-synth/  
  data/  
    raw/  
    bronze/  
    silver/  
    gold/  
  configs/  
  study/  
  concepts/  
  scenarios/  
  sql/  
    extract_t2dm_cohort.sql  
  src/  
    ingest/  
    normalize/  
    omop_lite/  
    cohorts/  
    features/  
    analysis/  
    economics/  
    reporting/  
    cli/  
  tests/  
  reports/  
  docs/  
  notebooks/
```

Represent eligibility, exclusions, index rules, baseline windows, follow-up windows, outcome definitions, and subgroup definitions in YAML or JSON config files. Expose the pipeline through a Typer CLI. Generate the poster-ready artifact through a parameterized Quarto report rather than notebook copy-paste.

The interface layer should also be concrete enough that an AI or collaborator could scaffold the repo immediately:

```
# src/extraction.py
class CohortExtractor:
    def __init__(self, raw_data_dir, db_path, config): ...
    def load_and_clean_patients(self): ...
    def apply_inclusion_criteria(self, patients_lf): ...
    def _log_pipeline_run(self, final_count, status): ...

# src/statistics.py
class CausalInferenceEngine:
    def __init__(self, analytical_df): ...
    def calculate_propensity_scores(self, covariates, treatment_col): ...
    def run_survival_analysis(self, time_col, event_col, treatment_col): ...

# src/economics.py
class MarkovDecisionEngine:
    def __init__(self, baseline_transition_matrix, drug_costs, state_utilities):
    ...
    def run_cohort_simulation(self, cycles, cohort_size, initial_state,
discount_rate): ...
```

For analytics, use lifelines for Kaplan-Meier, log-rank, and simple Cox modeling; keep statsmodels for straightforward count or GLM work; reserve more elaborate survival modeling for later. For testing, focus first on temporal correctness: baseline windows, exclusion logic, observation continuity, censoring, and event-date derivation.

The single best economics-specific test to preserve for the later phase is the Markov trace conservation check:

```
def test_markov_trace_preservation():
    for cycle_sum in np.sum(trace, axis=1):
        assert pytest.approx(cycle_sum, abs=1e-5) == 1000.0
```

That unit test matters because it proves population is conserved across Markov cycles, which is one of the few crisp technical signals that the economic layer is mathematically sane.

Your reproducibility layer should be methodologically visible. Fix the Synthea seed and reference date, preserve metadata export, hash raw inputs, version concept sets, and save a manifest for every analytic run. This is what moves the repo from student script to serious methods asset.

HEOR Realism and Research Pathway This project teaches real HEOR skills only if it is written and presented honestly. The real skills are translating a clinical question into a cohort protocol, defining index and observation windows, handling inclusion and exclusion logic, building baseline covariates, aligning exposures and outcomes temporally, summarizing treatment patterns and monitoring, doing simple survival and utilization analyses, structuring payer-perspective cost summaries, and documenting provenance, missingness, and limitations.

What it cannot honestly claim matters just as much. Synthetic data adoption is still limited by concerns about quality, representativeness, trust, and context of use. Synthea maintainers have openly noted that medications often function like placebos unless module logic explicitly links them to downstream outcomes. So you cannot present Phase 1 as if it estimated real comparative treatment effectiveness.

The strongest first poster is therefore a methods-and-pipeline poster with a disease-specific demonstration, not a pseudo-clinical conclusion. By the end of PY1, the evidence package should contain five concrete things before you call it submission-ready: a locked study protocol; reproducible cohort counts and a cohort diagram; a baseline table and treatment-pattern figure; one survival or utilization figure plus a cost summary; and a limitations section that explicitly states the data are synthetic and not a substitute for real retrospective RWD. If you have those, you have a real poster. If you do not, you have a code demo.

The poster framing should branch cleanly into two possible angles:

Clinical Outcomes Path

- Focus: T2DM treatment-pattern and outcome demonstration in a synthetic cohort
- Audience: managed care and applied HEOR reviewers
- Risk: reviewers may overread the clinical claim if the synthetic limitations are not foregrounded

Methodological Validation Path

- Focus: reproducible cohort extraction, OMOP-lite mapping, attrition logic, comparative-methods sandbox, and economic-model plumbing
- Audience: HEOR methods and informatics reviewers
- Strength: this is the safer and stronger Phase 1 poster angle

Right after the checklist, the abstract should be planned with a fixed component matrix:

Abstract Section Content Strategy for a Methodological Validation Poster	
Background	Access to retrospective RWD is a major barrier to HEOR training and algorithm development. Synthetic data provide a privacy-free sandbox, but still require rigorous mapping, study design, and transparent limitations.
Objective	Engineer an open-source, OMOP-lite, T2DM cohort extraction pipeline that produces reproducible cohort definitions, descriptive outcomes, and a clear pathway to later comparative and economic modeling.
Methods	Map raw Synthea CSVs into Parquet plus a lightweight SQLite relational layer; apply a prespecified incident T2DM cohort design with deterministic inclusion and exclusion logic; generate attrition, baseline, treatment-pattern, survival, utilization, and synthetic-cost outputs.
Results	Report the extracted cohort counts, the final attrition diagram, baseline balance table, treatment-pattern summary, and one survival or utilization result with synthetic-cost context.
Conclusions	Synthetic OMOP-aligned environments are effective, reproducible, zero-PHI sandboxes for validating HEOR data architectures, cohort definitions, and downstream analytic modules before IRB-gated real-world data access.

Conference-specific submission constraints should also be built into the drafting workflow rather than treated as an afterthought. AMCP Nexus requires abstracts of no more than 2,600 characters including spaces, uses a strict Background, Objective, Methods, Results, Conclusions structure, and explicitly rejects results in progress. That means the pipeline must already have final cohort counts, figures, and output language locked by the submission window. ISPOR is attractive not only because of its HEOR prestige but because it offers Best Student Poster Research Presentation Awards and provides specific taxonomy lanes that fit this repo well, especially Methodological & Statistical Research and Modeling, Simulation, Optimization.

Because your success marker is end of PY1, June 2027, the realistic target is a June 2027 internal abstract-ready milestone, then adaptation to whichever external call fits best.

Pharmacy-School and Fellowship Positioning When speaking to pharmacy faculty, the right framing is: "I'm building a reproducible synthetic RWE pipeline in type 2 diabetes to learn cohort construction, temporal study design, survival and utilization analysis, and payer-oriented evidence workflows in a no-PHI environment."

When speaking to research mentors, bridge it directly to your oncology work: "My oncology work is retrospective and externally constrained by fixed datasets; this project complements that by giving me a controlled environment to pre-specify cohort logic, outcome definitions, and analytic workflows before translating those habits to real-world data."

For early HEOR, managed care, or fellowship positioning, the stronger line is: "I used pharmacy school to build a payer-relevant evidence-generation portfolio from the ground up, starting with a synthetic RWE cohort pipeline in metabolic disease, then extending it toward comparative-methods training and payer-oriented economic modeling."

Risks, Roadmap, and Final Recommendation The biggest risk is performative sophistication: building something that looks advanced but teaches the wrong lessons. In practice that means rushing into causal inference without stable cohort logic, overstating synthetic findings, hard-coding disease logic into notebooks, or spending scarce time on dashboards before the extraction engine is defensible.

A second risk is economic overreach. Budget impact and cost-effectiveness are valuable, but only after the cohort and resource-use layer is defensible. If you jump to a model before your cohort definitions, event rates, and resource groups are stable, you will produce something that looks payer-flavored but is analytically hollow.

A third risk is architecture bloat. Do not start with a full enterprise OMOP conversion, a packaged web app, a plugin system, or a disease-agnostic platform framework. The right answer is not "no OMOP"; it is "OMOP-lite only." Map the core Synthea entities into a lightweight local relational layer because that is high-yield industry signaling, but stop well short of a full CDM implementation until the study definitions and cohort outputs are stable.

Your pre-matriculation plan should be Phase 1 only. Lock the disease area to T2DM. Write a two-page protocol with objective, index date, inclusion and exclusion criteria, baseline and follow-up windows, outcomes, and reporting language. Generate one fixed Synthea dataset with a documented seed and reference date. Build raw-ingest, Parquet-normalization, and

OMOP-lite SQLite mapping jobs. Implement the first version of the cohort DSL and produce an attrition table plus initial baseline table. If you finish this window with a stable cohort extractor, manifest logging, and reproducible cohort counts before classes begin, you are on track.

The next block should protect PY1 Session 1 from overreach. Use early-session gaps and short breaks to harden baseline covariates, medication class mapping, diabetes-related lab extraction, SQL extraction scripts, the Typer CLI, and the Quarto report. Do not require PSM or Cox modeling to be complete before school starts.

Phase 2 should begin in the two-week breaks between sessions and finish during Session 2. That is when you add propensity scoring, matching or IPTW, SMD diagnostics, Kaplan-Meier outputs, and any exploratory Cox modeling. By the end of that window, the repo should be something you could show a faculty member without apology: README written for a collaborator, one sensitivity analysis, one transparent synthetic-cost layer, and a frozen poster storyboard with background, objective, methods, attrition, baseline, outcomes, and limitations.

The final call is straightforward. Build one repo. Start with a T2DM cohort pipeline. Include an OMOP-lite SQLite mapping layer in Phase 1. Do not build PS matching before the deterministic cohort engine is stable. Do not build a dashboard yet. Do not make causal claims yet. Define Version 1 success as a documented, reproducible pipeline that produces a cohort, a baseline table, a treatment-pattern summary, one time-to-event or utilization analysis, one utilization and cost summary, and a poster-ready report. Move to later phases only after the cohort engine, reporting pipeline, and limitations language are stable. That is the strongest merged version of the three source documents: ChatGPT's scope discipline, Gemini Thinking's technical specificity, and Gemini Pro's OMOP framing and abstract structure.